

Instructions:

Prepare columns: Y=Total_Weekly_Sales; X=[Store_Size, Temperature, Total_MarkDowns, Unemployment
Data → Data Analysis → Regression; include headers; output below.

Paste output starting at A12.

Q11a – Model Performance Comparison

Metric	Simple Regression	Multiple Regression
R-squared	0.691666722	0.75023546
Adjusted R ²	0.690946317	0.747290124
Standard Error	367654.776	332456.14

The multiple regression model is better than the Simple Regression. The Adjusted R-squared value improved by .05, which tells us the additional variables we provided in the multiple regression explain more of the variance in weekly sales.

Q11a Response

Q11b – Overall Model Significance: Compare Significance F to $\alpha=0.05$ (interpret). The overall model is statistically significant, with a Significance F value very close to zero.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.866161336
R Square	0.75023546
Adjusted R Square	0.747290124
Standard Error	332456.14
Observations	430

ANOVA

	df	SS
Regression	5	1.40767E+14
Residual	424	4.68635E+13
Total	429	1.87631E+14

	Coefficients	Standard Error
Intercept	-1260185.689	156795.7021
Store_Size	8.910912813	0.304050525
Temperature	5763.395308	1112.060985
Total_MarkDowns	5.922780632	1.175775203
Unemployment	124496.346	16841.78484
IsHoliday	74010.2494	76438.07239

RESIDUAL OUTPUT

<i>Observation</i>	<i>Predicted Total_Weekly_Sales</i>	<i>Residuals</i>
1	1487757.065	153288.2052
2	1385863.508	73737.66176
3	1379275.89	15117.94989
4	1336564.252	-17238.66164
5	1781098.877	-144759.2267
6	1568803.847	233673.5826
7	1458180.025	361689.9746
8	1488527.862	50859.96793
9	1653284.942	35135.81837
10	1440359.22	235071.9398
11	1429691.729	247781.0512
12	1443461.64	67606.43037
13	1479206.333	170398.297
14	1492704.407	406972.4728
15	1448838.033	172193.6666
16	1414833.475	106744.3954
17	1424479.801	44448.56897
18	1588175.887	96344.10252
19	1485133.902	125962.1485
20	1462050.174	133851.6961
21	1480809.612	74634.93826
22	1540122.568	157108.3925
23	1501316.591	129290.4088
24	1538138.898	-10293.08826
25	1518623.517	21797.97336
26	1552881.732	216972.4279
27	1474428.769	52585.27076
28	1465143.834	32810.9257
29	1555664.654	-116540.9442
30	1707469.422	-76333.63241
31	1577484.141	14925.82887
32	1496460.159	101407.8908
33	1502101.708	-7979.327887
34	1636022.015	-53938.61541
35	1578049.734	83717.59589
36	1538730.49	-21301.61964
37	1432798.619	73327.44062
38	1444936.088	-7876.827505

39	1392417.873	278368.0968
40	1320592.565	252480.2446
41	1318053.146	190015.6239
42	1334609.607	159050.1332
43	1511485.703	22316.52717
44	2004204.997	-204684.8573
45	1813875.861	-69150.38138
46	1814408.205	-102639.0952
47	1748724.406	-87818.26608
48	2537556.053	-602256.1126
49	2026336.785	76985.89466
50	1909455.598	287232.862
51	1977613.518	-115810.8176
52	2001510.528	-48954.86832
53	1882945.224	54683.03579
54	1852838.473	123243.6575
55	1867445.468	-77006.30814
56	1919055.822	-61574.98182
57	1928562.224	200473.6865
58	1921449.193	14419.90681
59	1865168.244	-17823.79404
60	1822896.382	-58763.29176
61	2056996.847	-133039.7572
62	1976652.043	-59131.0529
63	1915525.496	85415.17356
64	2030245.842	-117454.752
65	1981740.028	-71647.65767
66	2015035.162	-4818.672446
67	1974403.805	-11479.50497
68	2193217.577	-305484.3668
69	1995043.719	-113997.5994
70	1993087.769	48419.63119
71	1936850.431	-106775.301
72	1917045.207	-97378.74746
73	1967381.148	-209457.2683
74	2213376.879	-267272.2387
75	1960152.365	-93432.40481
76	1933223.28	-5207.269983
77	1880444.702	-3656.551761
78	2044819.292	-97735.99215
79	2040120.396	-141343.3255
80	1912613.502	-97806.87221
81	1860428.365	-31012.69516
82	1892429.728	-145959.1678

83	1792263.676	206057.3637
84	1740192.651	160552.4788
85	1754508.047	93482.36284
86	1770249.572	64208.77825
87	315372.6711	82805.53886
88	277095.0424	90343.57759
89	266355.4904	99463.11956
90	266948.4577	82569.6423
91	378814.0505	46146.60947
92	384268.9023	89023.56771
93	284069.8712	191521.2088
94	330385.0764	88540.39364
95	328433.5607	141318.9993
96	341257.3635	103904.6865
97	316679.8629	95095.9371
98	315725.0887	98182.16131
99	331900.458	75588.38198
100	363734.698	139497.432
101	376236.3488	44553.3912
102	343063.5074	91758.62262
103	334929.8703	59686.23968
104	390832.4937	49081.0763
105	366965.4267	65019.93329
106	344432.2591	73680.50089
107	381844.8329	31856.45707
108	404961.5976	27306.93237
109	421840.9831	24495.81687
110	411395.2072	30679.58276
111	403099.6108	16398.33921
112	423726.3028	-760.9727836
113	372427.6466	38778.85344
114	359315.0431	57598.05693
115	342704.6018	89720.24816
116	388903.5844	524.3155912
117	412775.3755	7214.914541
118	390962.7658	848.8341966
119	376115.6877	18803.14227
120	371846.8491	40602.82094
121	404763.7335	4074.996473
122	446976.3091	-38746.57912
123	357222.9948	50366.16516
124	314931.2074	99460.88257
125	332295.8443	57517.17573
126	260514.2501	183043.3999

127	238951.2576	171853.1324
128	269531.5214	154981.5586
129	257877.4977	147555.2023
130	1704038.761	343727.309
131	1494608.655	447067.9547
132	1456883.13	548214.6295
133	1427676.838	501043.6722
134	2044308.689	129065.2211
135	1661881.934	712778.7063
136	1610018.361	817621.8085
137	1534063.242	692598.9283
138	1641307.986	565011.9144
139	1575098.35	627352.4596
140	1591961.056	623006.3838
141	1570794.068	520798.4723
142	1611281.325	478100.4455
143	1648258.509	821947.6212
144	1665114.443	440186.947
145	1589164.21	555172.6798
146	1599740.661	464324.9988
147	1722695.978	474272.3523
148	1675189.425	452471.7449
149	1572735.63	634479.1803
150	1631963.131	522174.5386
151	1662108.691	517252.2493
152	1688759.324	556497.8562
153	1721379.116	512811.8142
154	1718948.595	478351.0555
155	1669779.699	458583.2214
156	1690557.905	533941.375
157	1642181.853	458070.7568
158	1598109.814	577453.8757
159	1694055.199	354558.4514
160	1937185.938	237328.1915
161	1714241.885	479125.8053
162	1594545.029	688995.271
163	1668185.002	457056.6781
164	1719784.821	361396.5294
165	1773468.34	351636.3804
166	1652479.646	465374.9537
167	1609801.978	509636.5523
168	1596126.205	431494.0253
169	1518542.88	691292.5501
170	1460080.16	672945.9098

171	1515351.771	581915.0794
172	1480186.84	669407.6201
173	171646.7653	140431.9447
174	124902.3401	166552.1799
175	130729.1591	156794.8209
176	119631.4567	176342.7633
177	377450.0579	-43502.05788
178	226636.1428	122603.7372
179	145496.6505	210931.3295
180	161387.8838	150832.5862
181	202698.4512	156507.7588
182	164703.01	182592.59
183	181976.8242	157415.7158
184	172034.9016	149265.0884
185	223864.7815	107453.9485
186	231447.7068	171537.9932
187	204144.3651	147687.6649
188	179632.5886	150430.4714
189	192525.8056	132313.9344
190	279200.6543	81732.03566
191	238517.5097	95353.01031
192	208832.0692	127357.5908
193	263230.5342	78763.94579
194	264165.9976	95701.80245
195	275075.7967	66628.79335
196	268840.8586	58542.78144
197	269693.5867	55348.09327
198	289350.1246	40307.97543
199	260244.7143	80969.71568
200	232487.6596	83715.98041
201	228944.6768	92260.44321
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205	280988.2017	33025.97828
206	229969.2402	90862.1198
207	297746.3395	46895.67053
208	354156.1185	-3507.208467
209	230919.1331	68881.53686
210	203478.1365	103828.6235
211	218526.3085	91615.37148
212	166861.6855	176186.6045
213	139211.0008	186134.4092
214	133843.7892	179514.3608

215	152349.2584	167201.5116
216	1855431.361	-460091.651
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218	1698068.629	-371812.929
219	1647316.991	-331706.331
220	2077912.519	-581606.7391
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222	1840413.025	-207796.9348
223	1792489.183	-327301.4726
224	1865834.723	-315449.0735
225	1726407.034	-157102.6341
226	1737404.448	10605.84173
227	1731955.972	-236812.3522
228	1802663.464	-293730.2039
229	1803616.524	36514.66594
230	1792688.948	-176294.498
231	1733811.196	-277737.9557
232	1729386.617	-273165.5168
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234	1846577.975	-329502.3049
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236	1826865.301	-223071.8806
237	1833928.997	-152807.6167
238	1860676.392	-164056.8716
239	1856927.112	-214679.6322
240	1858786.329	-240514.0793
241	1841625.487	-192762.0274
242	1822323.096	54036.29449
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244	1807074.64	-232712.6695
245	1908532.858	-395303.6982
246	2073905.708	-446630.7785
247	1854478.084	-266097.3537
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250	1978104.998	-400665.1877
251	1944582.699	-336505.6888
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253	1768787.935	-343184.2852
254	1749356.636	-380225.1764
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256	1651108.192	-191711.3525
257	1642469.995	-205586.0054
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261	610658.8187	-32655.96865
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266	661362.7467	-58321.60667
267	884774.9613	-333716.8313
268	669233.5827	-90067.08275
269	657433.3491	11772.3809
270	660542.7529	-44545.46291
271	719396.9986	-136074.7986
272	704803.3944	-83377.4144
273	647793.7535	-130373.5235
274	633523.9358	-176183.8758
275	686451.0307	-218623.2807
276	693138.5816	-227939.6916
277	815869.8652	-355472.4552
278	695376.1515	-226947.8515
279	715739.1241	-182999.3541
280	729906.7643	-131411.7443
281	786482.9799	-143519.2299
282	829561.2912	-162619.2712
283	836175.1417	-148830.4617
284	791121.2639	-86785.85391
285	762796.2498	42846.36019
286	1070090.714	-375939.8235
287	735587.4635	-49241.7735
288	752067.1548	-65701.75484
289	823552.2282	-142597.4182
290	758064.7021	-82138.40207
291	719987.5871	-77537.18715
292	739532.4974	-130433.1274
293	757523.9886	-171056.8286
294	805045.0497	-207168.4997
295	714186.3618	-173066.1618
296	702625.2773	-171783.0273
297	680069.6047	-154523.8447
298	628553.964	-122723.404
299	556613.5837	-53149.65368
300	597115.0462	-80690.21615
301	581058.7851	-85515.50509
302	1299753.427	-427640.1967

303	1197144.705	-379482.9453
304	1153879.774	-339924.9544
305	1293332.205	-515153.6748
306	1498181.248	-570570.5583
307	1269170.367	-247769.9466
308	1186117.103	-89884.2132
309	1246752.985	-318215.4448
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311	1210875.869	-250760.309
312	1219197.788	-298019.3985
313	1209869.802	-335646.5521
314	1280206.574	-374271.2837
315	1261579.067	-214762.4774
316	1293188.232	-383198.7823
317	1328029.504	-455741.044
318	1286008.168	-406559.9175
319	1365162.54	-427930.4502
320	1290927.03	-370798.1397
321	1273913.359	-359991.3493
322	1355538.805	-460381.3653
323	1369139.09	-447977.8897
324	1344745.878	-415925.878
325	1367237.731	-450319.0311
326	1346180.487	-446730.8366
327	1402545.77	-524247.5496
328	1368904.759	-432699.2589
329	1330639.28	-440151.2703
330	1642191.178	-753357.108
331	1342699.132	-504471.6119
332	1417663.172	-520586.4419
333	1376912.024	-446166.3343
334	1317589.002	-420975.8116
335	1267641.094	-331267.4441
336	1331834.616	-355696.8862
337	1426569.418	-494409.0484
338	1248092.147	-364522.7667
339	1244616.941	-386820.4912
340	1255252.184	-370527.774
341	1156303.112	-179867.092
342	1111023.391	-183511.401
343	1152743.529	-252433.7787
344	1161059.092	-269387.6521
345	932229.3931	-412643.7231
346	877770.0994	-402805.4994

347	907273.5766	-427143.5366
348	873039.9342	-390588.7242
349	1076989.237	-527021.3469
350	991833.0475	-382096.9275
351	929058.7294	-270093.6794
352	969015.5365	-405436.7465
353	966524.8635	-347026.5835
354	924114.7126	-349158.7626
355	1009125.915	-458752.3447
356	920640.9766	-369849.6566
357	992160.1054	-417174.7354
358	1034053.335	-356167.3449
359	990283.466	-411743.606
360	957311.7439	-414492.7139
361	1016816.499	-466401.5089
362	1050906.654	-464617.5737
363	996046.3702	-403474.0702
364	982881.4088	-411417.4788
365	1053229.056	-506003.056
366	1051413.966	-467765.3764
367	1060594.568	-478069.1481
368	1074917.363	-510311.263
369	1069377.942	-507204.8217
370	1082857.47	-538086.7695
371	1044515.942	-465725.5818
372	1038140.794	-501603.1545
373	1022148.987	-508157.4173
374	1046354.859	-550403.8594
375	1131449.591	-597562.0509
376	1074057.37	-535343.9002
377	1027550.78	-492397.3097
378	1004168.367	-431280.5868
379	1045354.681	-468475.5305
380	1139337.173	-573524.8827
381	985279.3811	-461852.0311
382	961258.2135	-427501.3335
383	982680.5875	-466319.5275
384	893317.8865	-286562.5865
385	844116.1445	-285651.3445
386	891629.5103	-349620.0503
387	891495.9224	-341764.4324
388	1459085.59	224316.1899
389	1253730.99	457831.7401
390	1193106.319	482456.6208

391	1154587.739	477818.2613
392	1629172.816	238230.1941
393	1331222.016	887373.7836
394	1404296.241	764413.5185
395	1281371.357	758044.3825
396	1325293.532	665077.4875
397	1225693.167	691789.9329
398	1212862.226	717952.4336
399	1172121.917	665335.7733
400	1269040.687	546719.7329
401	1300575.651	862808.5186
402	1327277.997	647409.5132
403	1266369.86	510796.6696
404	1286267.297	426720.2631
405	1383432.667	437931.7532
406	1333509.617	458835.6833
407	1357712.286	437440.4444
408	1361460.429	469478.6707
409	1343907.772	423563.7085
410	1367345.868	473145.5424
411	1370038.796	441524.0843
412	1412723.327	342610.8528
413	1361246.065	346235.8355
414	1397691.637	408308.1531
415	1375908.847	389663.0635
416	1374236.084	495730.9456
417	1415316.834	402286.8259
418	1612079.246	327360.8443
419	1494470.285	385966.6546
420	1337911.444	489885.9557
421	1384616.501	380367.6488
422	1511402.445	138883.0955
423	1419171.146	289112.1336
424	1348070.751	292098.2394
425	1365088.14	306769.43
426	1302077.766	392784.6437
427	1350218.676	408752.7044
428	1241926.058	471963.0518
429	1190824.74	544010.0804
430	1212367.683	531981.3666

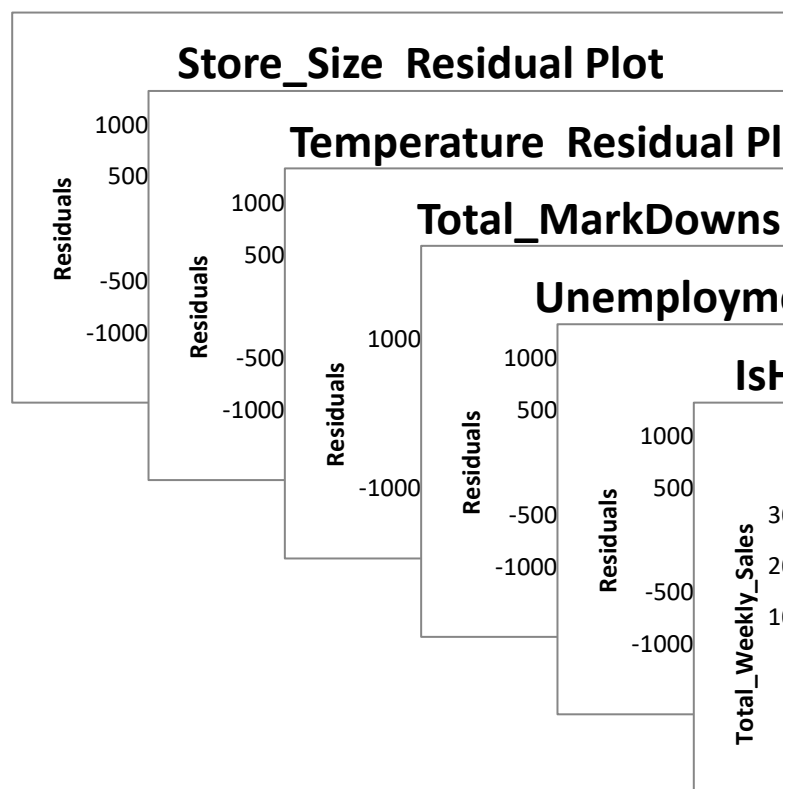
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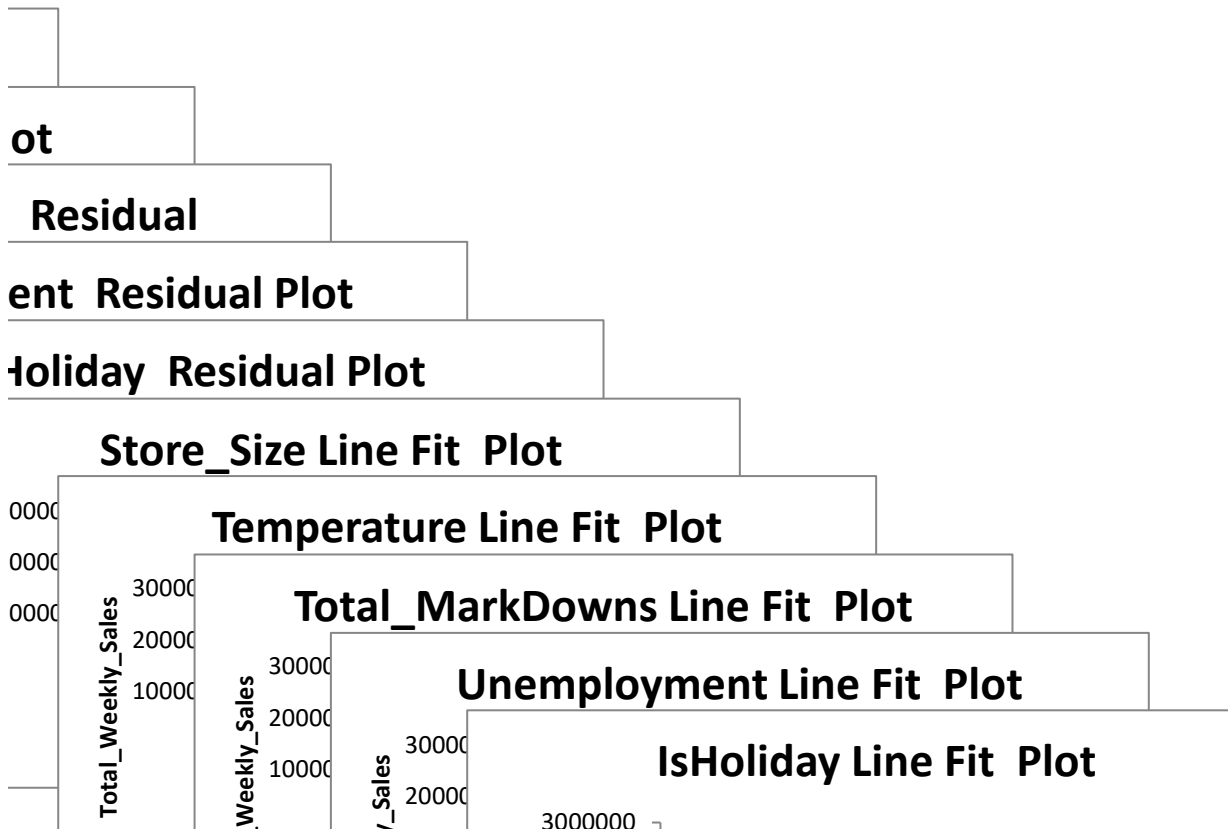
Improvement
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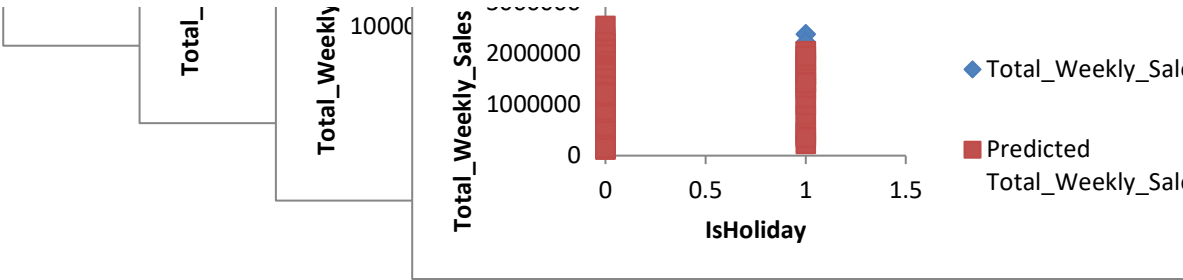
<i>MS</i>	<i>F</i>	<i>Significance F</i>
2.81534E+13	254.7197736	2.8843E-125
1.10527E+11		

<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>
-8.037118823	9.23081E-15	-1568379.354
29.30734229	5.4683E-104	8.313278794
5.182625222	3.39193E-07	3577.556374
5.037340995	6.99828E-07	3.611706643
7.392111177	7.77459E-13	91392.55981
0.96823804	0.333477466	-76234.49161

Upper 95% Lower 95.0% Upper 95.0%		
-951992	-1568379	-951992
9.508547	8.313279	9.508547
7949.234	3577.556	7949.234
8.233855	3.611707	8.233855
157600.1	91392.56	157600.1
224255	-76234.5	224255







7

es

es